Format WPR

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 11) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 01 2. 02 3. 03
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Synthesized academic benchmarks for enterprise scale: Jayavibhav Prompt Injection (327K), Stanford SPML (16K), PKU-Alignment BeaverTails (330K), Deepset, LibrAI Do-Not-Answer, XSTest v2 (exaggerated safety), FEVER Gold Evidence (NAACL 2018), and HaluEval QA (EMNLP 2023). Conducted comparative literature survey against Meta Llama-Guard-3 (8B) and SelfCheckGPT multi-sample hallucination checking.
Technical & Economical Feasibility	Feasible. Dual-tier hardware architecture formulated: Edge workstation deployment on NVIDIA RTX GPU (4GB–20GB VRAM) and Enterprise Microservice deployment on high-VRAM GPU (47GB VRAM). Tested with CUDA acceleration, PyTorch 2.6, and Transformers. Zero monetary expenditure (₹0).
Bill of Material	₹0.00 (Fully open-source software stack, public academic benchmark datasets, and local developer environment)
Project Progress Schedule (PERT Chart)	• W1-W5: Problem formulation, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, NLI fact verifier, and multi-model benchmarking (Completed) • W9-W10: Stage 2 authentic benchmark integration (FEVER + HaluEval), NLI declarative statements, 95% Wilson CIs, and E2E latency measurement (Completed) • W11: Interactive unified pipeline with multi-model backbone switching, negative refutation hallucination resolution, 15k massive multi-domain dataset curation, high-VRAM DeBERTa-v3-large trainer (512-token context), and Meta Llama-Guard-3 enterprise baseline harness with LaTeX table generation (Completed / Week

	11) • W12: Final IEEE/ACM conference research paper manuscript compilation, camera-ready Overleaf integration, and final viva defense preparation.
Design of critical components	1. Interactive Real-Time Sidecar Shell (`src/guardshield_unified_pipeline.py`) 2. Negative Refutation Fact Verifier & Multi-Source Grounder (`src/nli_verifier.py`) 3. Massive Multi-Domain Dataset Curation Pipeline (`src/prepare_massive_training_data.py`) 4. High-VRAM DeBERTa-v3-Large Training & Evaluation Suite (`src/train_prescan_large.py`) 5. Enterprise Baseline Benchmarking Harness & LaTeX Exporter (`src/benchmark_enterprise_baselines.py`)
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	98% (Unified end-to-end pipeline operational; massive 15k dataset splits created; high-VRAM large trainer verified; Meta Llama-Guard-3 and SelfCheckGPT baseline harnesses operational with automated LaTeX table exports)
Result and Analysis	Ingress Pre-Scan achieves 98.42% in-distribution F1 (0.87% FPR, 4.02ms latency) and 0.0% False Positive Rate on XSTest exaggerated-safety prompts (vs 4.8%–5.0% for Meta Llama-Guard-3-8B). DeBERTa-v3-large supports 512–1024 token contexts with 4.95ms latency. GuardShield is ~18× faster than Meta Llama-Guard-3 (14.2ms vs 239.8ms) and ~220× faster than SelfCheckGPT (14.2ms vs 3,073.3ms) with 11× lower peak VRAM (1,420 MB vs 16,384 MB). Negative refutation parsing resolves 100% of false-positive suppressions on valid negative assertions.
Report writing	WPR Week 11 drafted, verified with camera-ready LaTeX tables, dataset manifests, and formatted.
Signature of students	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh

Work done in this week (Week 11):

1. Interactive Unified Real-Time Sidecar Shell & Multi-Backbone Orchestration: Built the unified end-to-end interactive CLI pipeline (`src/guardshield_unified_pipeline.py`) seamlessly connecting Stage 1 Ingress Pre-Scan -> Real-Time Streaming Token Logit Shannon Entropy H(x) with Circuit Breakers -> Stage 2 Declarative NLI Fact Grounding. Integrated a dynamic model-switching engine supporting Qwen2.5-0.5B-Instruct, GPT-2 Base (124M), Phi-3.5-mini-instruct (3.8B), TinyLlama-1.1B-Chat-v1.0, and arbitrary HuggingFace repository IDs with robust alias resolution and crash recovery.

2. Resolution of Negative Refutation Hallucination False-Positives: Diagnosed and resolved the critical encyclopedia retrieval and NLI edge-case bug where valid negative claims (e.g., "There is no capital of Mars as it is a planet, not a country") were misclassified as hallucinations. Implemented negative refutation semantic parsing (`is_refutation`), recognizing valid refutations as FACTUALLY_CONSISTENT_REFUTATION. Upgraded Wikipedia grounding with multi-source search query ranking, multi-word named entity preservation ("World Wide Web", "Canberra"), and strict contradiction-only egress withholding (P(contradiction) >= 50%).

3. Massive Multi-Domain Benchmark Curation Pipeline: Developed `src/prepare_massive_training_data.py`, automating the streaming ingestion, deduplication, and balancing of 15,304 benchmark samples across public academic datasets: Jayavibhav Prompt Injection (327K corpus), Stanford SPML (16K corpus), PKU-Alignment BeaverTails (330K safety corpus), Deepset, LibrAI Do-Not-Answer, FEVER NAACL 2018, and HaluEval EMNLP 2023. Exported dedicated camera-ready holdouts for in-distribution test (986), XSTest v2 exaggerated safety (450), long-context prompts (500), FEVER (500), and HaluEval (500).

4. High-VRAM Pre-Scan Training Architecture for Enterprise Hardware: Engineered `src/train_prescan_large.py` tailored for 20 GB VRAM (Device 1) and 47 GB VRAM (Device 2). Scaled model backbone to `microsoft/deberta-v3-large` (435M parameters, 5x capacity increase) with extended sequence context (512–1024 tokens), non-reentrant gradient checkpointing, and BF16/FP16 mixed precision. Integrated automated post-training evaluation across in-

distribution, XSTest (FPR), and long-context splits with hardware latency and peak CUDA VRAM tracking. 5. Enterprise Baseline Benchmarking Harness & Camera-Ready LaTeX Generation: Built `src/benchmark_enterprise_baselines.py` for Device 2 (47 GB VRAM) executing head-to-head empirical comparisons between GuardShield AI, Meta Llama-Guard-3 (8B), and SelfCheckGPT (5x stochastic sampling). Established that GuardShield is ~18x faster than Meta Llama-Guard-3 (14.2ms vs 239.8ms) and consumes 11x less VRAM (1,420 MB vs 16,384 MB) while reducing false-positive over-refusals from 4.8% to 0.0% on XSTest benign triggers. Automated generation of camera-ready LaTeX code for Table 1 and Table 2 for the IEEE conference paper. 6. Dual-Device Research Scaling Architecture Plan: Formulated comprehensive scaling blueprints (`scaling_vram_plan.md` and `research_scaling_and_benchmarking_plan.md`) defining exact experimental variable segregation between Device 1 (20 GB VRAM) for large encoder training and context-length ablations, and Device 2 (47 GB VRAM) for heavy 8B baselines and 50-client vLLM concurrent microservice stress testing.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date Signature of Guide